

concepts to cinema sound: audio objects and overhead speakers. Sounds of the onscreen story move all around in a 3D space, so you feel like you are a part of the action. This gives artists amazing capabilities to tell their stories, visualise games or perform music.

Many soundbars offer multichannel audio, but a few newer models also support two new surround sound formats: Dolby Atmos and/or DTS:X (Digital Theatre Systems). Both are object-based audio technologies, where sound engineers can place sounds almost anywhere in a listener's environment during the recording and mixing process.

Soundbars can create overhead sound effects, such as a plane flying above and then disappearing in the distance.

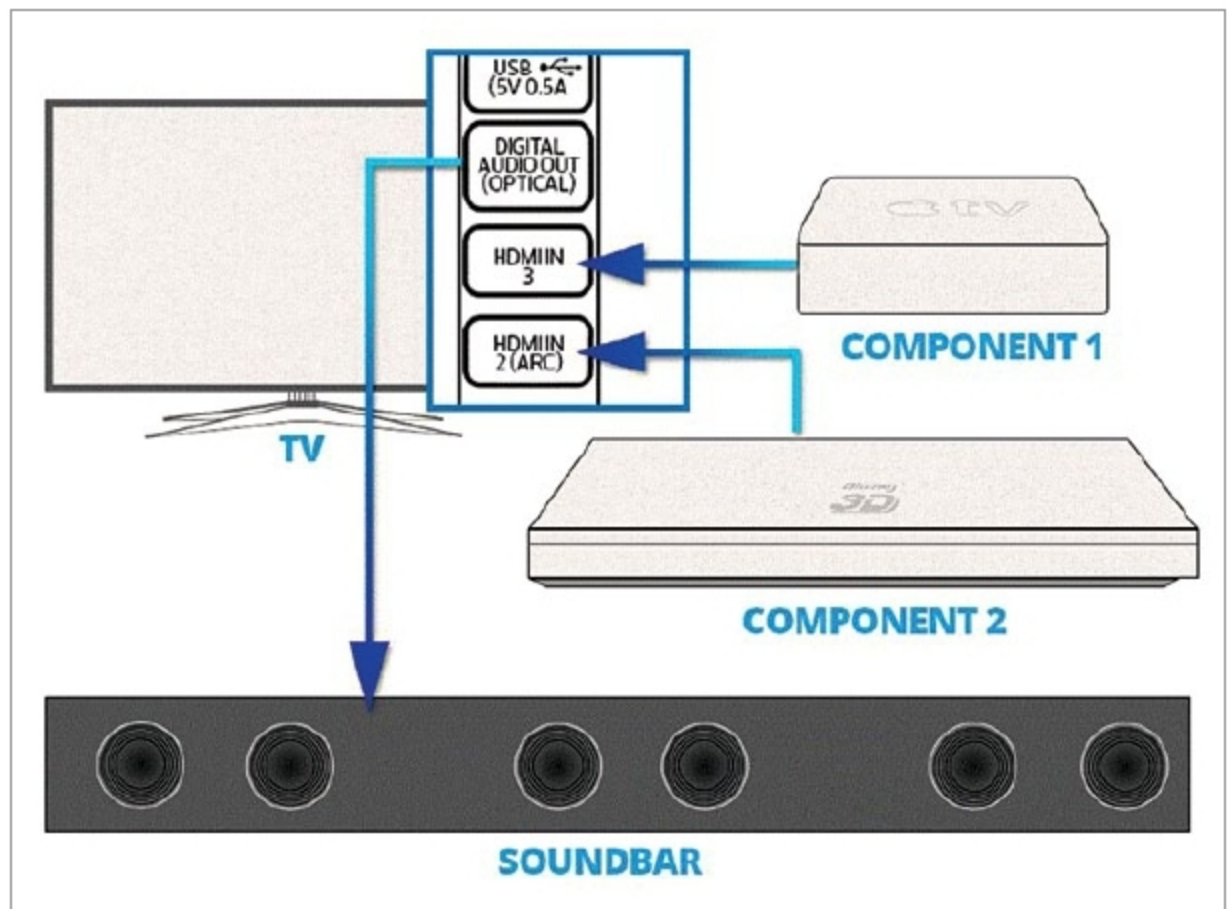
DTS:X is a bit different, in that it does not require special height speakers, although it will work with them. Instead, it remaps the sound effects based on the number of channels you have in your system.

Dolby Atmos soundbars work best in rooms with flat ceilings up to 3.35m (11-feet) high, which are made of hard, reflective material. The more the channels, the better and more immersive the experience.

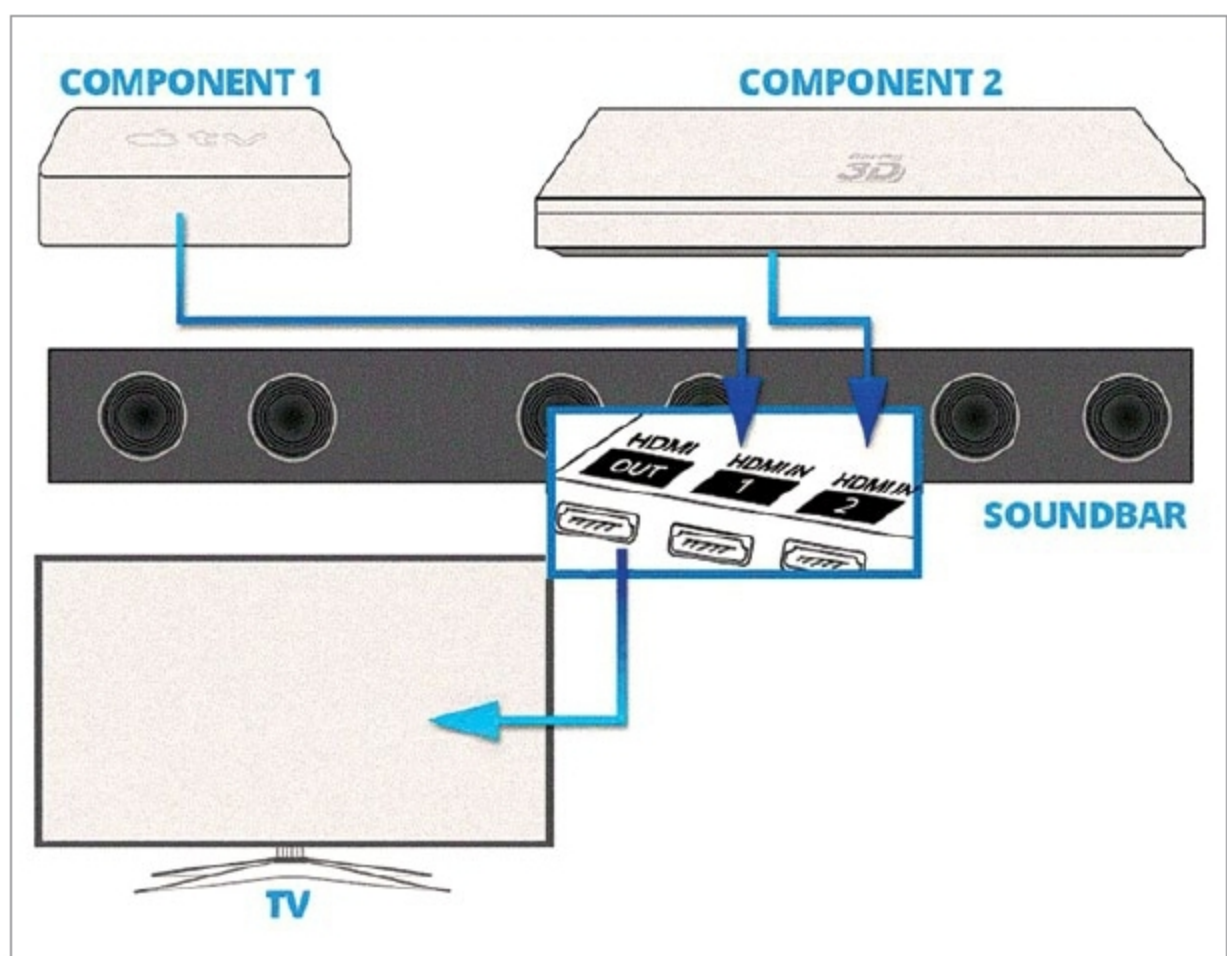
How soundbars work

Soundbars provide multi-directional audio for a home theatre experience. Sub-woofers are speaker drivers dedicated to the reproduction of low-frequency audio like exploding bombs, noise of a helicopter's blades, etc. Soundbars with a sub-woofer create a fuller sound, deep bass and more effectively project audio throughout the room. If you plan on watching lots of action movies or movies with epic music, a sub-woofer can improve your experience.

Stereo sound is divided between two channels—one speaker each on the left and right sides. Most TV programmes are recorded with this type of sound. Some newer soundbars come with Dolby Atmos technology that bounces the audio off ceilings to simulate a surround sound effect. Soundbars with upward-firing speakers reflect the sound above



Here, the TV has control of input selection and passes audio signal to the soundbar
(Credit: www.crutchfield.com)



Here, the soundbar keeps control of the system's inputs and passes video signal to the TV
(Credit: www.crutchfield.com)

and behind the viewer for a heightened experience, while front-firing speakers virtualise height in the 3D soundstage.

Meridian audio technology enhances sounds using digital signal processing (DSP). All noise, no matter how faint, can be heard. With differ-

ent available sound modes, audio experience can be matched to the entertainment onscreen.

Soundbars with virtual surround sound simulate the speakers that would be placed behind the sofa in a traditional 5.1 setup, without the need for additional speakers. Sound